

How to choose the right COB LED package

Let us now discuss some of the parameters that are important while choosing COB LEDs:

Forward voltage. This is the amount of volts the LED requires to conduct electricity and light up. A typical 20-watt COB LED would require a forward voltage of about 29V DC to 32V DC.

Forward current. It is the current flowing through an LED when forward voltage is applied to it. A typical 20-watt COB LED draws about 640mA. Therefore you should choose appropriate current driver as per the rating on the LED package.

Power consumption. An LED lamp packaging may be specified with power consumption in watts, lumen output, colour temperature, operating temperature range and others. Power rating in LED lamps should not be confused with power ratings in other lighting systems. With less wattage rating in LED lamps, as compared to other lighting products, you can get more light output or lumen output as shown in the table on the previous page.

Luminous flux. It is used as an objective measure of the total amount of visible light that your lighting source produces. The SI unit of luminous flux is lumen (lm). If you compare a CFL and a normal white LED, the latter is highly energy efficient and uses very little energy. The lumens produced by an LED bulb as compared to other lighting sources are shown in the table. COB LED offers a greatly improved lumen-per-watt ratio compared to other LED technologies.

Efficacy. This parameter is for deciding the performance of the LED. It is the effectiveness of the LED at converting electricity into visible light. Thus, efficacy is total

lumens per watt. For example, if a nine-watt light bulb comes with lumens of 800, it means it has an efficacy of about 90 lumens per watt.

CRI, or colour rendering index. CRI is a measure of light quality, showing how well a light source replicates colour to the human eye. White light is made up of all the colours in the light spectrum. We can see the colour of every material object in daylight. Natural

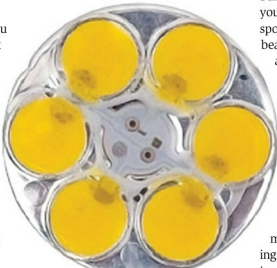


Fig. 4: MCOB LEDs

daylight has the highest CRI rating, while artificial light sources cannot produce all the colours of the light spectrum. CRI of 80 is the standard requirement for general home use.

CCT, or correlated colour temperature. CCT is measured in Kelvin (K) to refer to the warmth or coolness of the white light that the LED produces. The higher the temperature, the cooler the light looks. An LED bulb marked with 5000K does not actually heat up to 5000°F, but still gives off a warm light.

Some LED bulbs have more than 6000K CCT, which makes the light more towards bluish rather than white. A warm light is around 2700K, while neutral white at around 4000K and cool white at 5000K or more.

Lifespan of COB LED. Another important specification is the lifespan in hours. Some LED lamps may last for 50,000 hours. So if you consider daily use of eight hours, it may last 17 years.

Life of the LED also depends on the heat-sink used. COB technology gives the best conditions for optimal cooling, which, in turn, increases efficiency and lengthens the overall life of the lamp.

Beam angle. Beam angle of a bulb refers to the spread of light to your surroundings. For example, a spotlight in a kitchen will have a beam angle of between 35 degrees and 60 degrees. Standard and candle-shape bulbs require a beam angle as wide as possible when fitted to lamps. Though, ideally the beam angle should be over 200 degrees.

To distribute light better, to more areas, you would need a wider beam angle. But a narrow beam angle is required when you want to have the light more focused and intense. Adjusting beam angles correctly can vastly improve the lighting effect in a room.

Junction temperature. It is the highest operating temperature of the actual semiconductor in an electronic device. This temperature is normally specified in the datasheet. You will require this value when calculating the case-to-ambient thermal resistance for a given power dissipation. This will help you select an appropriate heat-sink.

COB LEDs are going strong

Many publicly-available industry reports claim that there is an increasing demand for COB LEDs in the global lighting industry. As of now, you can see COB LEDs in many consumer and industrial products including bulbs, downlights, tubelights, street lamps, bay lights and others. One can confidently state that their future looks bright, for now. **EFY**